

A contamination-robust search for infrared-excess technosignatures around white dwarfs: an empirical search of the 100 pc sample and projected sensitivity

[Author list]

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Abstract

Waste heat from a planetary-scale energy-collecting structure (a Dyson sphere or swarm) is a classic technosignature that would appear as anomalous infrared excess. Systematic photometric searches have so far targeted main-sequence stars, most prominently Project Hephaisostos, whose candidates proved vulnerable to background contamination. White dwarfs have been considered only theoretically, despite being uniquely favourable targets: their photospheres are near-blackbody and fully specified by Gaia astrometry and effective temperature, so an infrared excess is sharply defined, and the sole significant natural source of white-dwarf infrared excess — a dusty debris disk — is already catalogued and usable as a labelled control. We present the design, an open implementation, and the projected sensitivity of the first systematic catalogue search for infrared-excess technosignatures around white dwarfs, combining the Gaia EDR3 white-dwarf catalogue ($\sim 359\,000$ high-confidence white dwarfs) with CatWISE2020, unWISE, AllWISE and 2MASS photometry. Our central methodological contribution is a contamination-robust vetting funnel whose decisive, novel cut is a Gaia–WISE *co-movement* test: it requires the infrared source to share the white dwarf’s large proper motion, rejecting the chance-aligned background interlopers that defeated earlier broadband searches. At the CatWISE2020 source density a ~ 3 arcsec match implies a $\sim 10\%$ chance-alignment rate per white dwarf; the co-movement cut removes 45% of these (a factor of 1.8), with near-complete rejection for fast-moving white dwarfs. Using a realistic population model calibrated to the published Gaia and WISE statistics and end-to-end injection–recovery, we forecast that a null search of the 15 001 white dwarfs within 100 pc — of which $\sim 63\%$ (9 508) are detected by CatWISE2020 — would constrain the occurrence of warm ($T_{\text{dust}} \sim 500\text{--}1000$ K), substantial ($\tau \gtrsim 0.1$) Dyson-swarm-like excess to $f < 3.2 \times 10^{-4}$ at 95% confidence, the first such limit for the white-dwarf population. Applying the pipeline to the real 100 pc sample, we search 4 111 white dwarfs with CatWISE2020 and 2MASS or Gaia photometry, recover 155 with infrared excess (the warm-dust debris-disk population), and place an empirical 95% occurrence-rate limit of $f < 5.9 \times 10^{-3}$ after debris-disk subtraction, with 16 cool-excess candidates remaining for follow-up. The empirical limit is an order of magnitude weaker than the forecast, set by the near-infrared/Gaia anchoring fraction. We emphasise that broadband $W1/W2$ photometry cannot uniquely distinguish a Dyson swarm from warm dust or an unresolved brown-dwarf companion — the latter the dominant astrophysical false positive at the candidate level; surviving sources are prioritised follow-up targets, not detections.

1 Introduction

A sufficiently energy-intensive technological civilisation may surround its host star, in whole or part, with energy-collecting structures [a Dyson sphere or swarm; Dyson, 1960]. Thermodynamics

requires that the intercepted starlight be re-radiated as waste heat, so such a structure betrays itself through anomalous mid-infrared emission superimposed on (or replacing) the stellar spectral energy distribution [SED; Wright et al., 2014]. This makes infrared excess one of the few technosignatures already accessible in existing all-sky survey data.

The most ambitious recent search, Project Hephaistos [Suazo et al., 2024], cross-matched Gaia DR3, 2MASS and AllWISE for five million main-sequence stars and identified seven partial-Dyson-sphere candidates. Those candidates were, however, subsequently shown to be consistent with background contamination — blended or chance-aligned infrared sources — rather than genuine circumstellar excess [Tang et al., 2024]. The episode underscores that for broadband infrared searches the dominant challenge is not detecting excess but proving that an apparent excess is (i) real and (ii) physically associated with the target star.

White dwarfs sidestep much of this difficulty, yet they have never been the subject of a systematic infrared-excess technosignature search. Two properties make them exceptionally clean targets. First, a white-dwarf photosphere is essentially a blackbody whose absolute SED is fixed by the effective temperature and radius; with a Gaia parallax and a photometric or spectroscopic T_{eff} , the expected $W1/W2$ flux is predicted to ~ 5 per cent (our adopted systematic floor), so an excess is sharply defined. Second, the only astrophysically common source of white-dwarf infrared excess is a dusty debris disk produced by tidally disrupted planetesimals [e.g. Debes et al., 2011, Denny et al., 2020, Xu et al., 2021, Madurga Favieres et al., 2024, Murillo-Ojeda et al., 2026], and those disks are already catalogued. Rather than a nuisance, this single, well-characterised confounder is an asset: it provides a labelled control population that can be *subtracted*, leaving any residual anomalies as technosignature candidates. White dwarfs are also of independent interest as hosts: a post-main-sequence civilisation might track its cooling star inward, and the compact, low-luminosity white dwarf admits cooler, more compact structures [Zackrisson et al., 2022, Dyson H–R authors et al., 2026].

In this paper we (i) set out the design of a contamination-robust white-dwarf infrared-excess search, (ii) introduce and validate a Gaia–WISE co-movement cut that directly targets the Hephaistos failure mode, (iii) release a fully reproducible open-source pipeline, (iv) forecast the occurrence-rate sensitivity, and (v) carry out the search empirically on the 100 pc white-dwarf sample. Section 2 describes the input catalogues; Section 3 the SED prediction, excess statistic, contamination funnel and dust/technosignature discriminator; Section 4 the injection–recovery validation; Section 5 the projected sensitivity; Section 6 the empirical search and its results; and Section 7 the interpretation, limitations and outlook.

2 Data

Parent sample. We adopt the Gaia EDR3 white-dwarf catalogue of Gentile Fusillo et al. [2021], selecting the $\sim 359\,000$ high-confidence white dwarfs ($P_{\text{WD}} > 0.75$, tightened to 0.9 for the search), each with a photometric T_{eff} , $\log g$ and atmosphere type. The $\sim 15\,001$ white dwarfs within 100 pc form a nearly complete, well-characterised subsample that anchors our sensitivity forecast.

Infrared photometry. The primary infrared photometry is CatWISE2020 [Marocco et al., 2021, $W1$, $W2$, with proper motions], which is deeper than AllWISE (5σ limits $W1 = 17.43$, $W2 = 16.47$, Vega) and provides the proper motions our co-movement test requires. We use unWISE [Schlafly et al., 2019] forced photometry as an independent reduction to confirm that excesses survive de-blending, AllWISE [Cutri et al., 2013] for its quality flags and the neighbourhood multiplicity used in crowding rejection, and 2MASS for the near-infrared anchor. All access is catalogue-scale

through Virtual Observatory TAP services and the CDS X-Match service.

Control debris-disk catalogues. We assemble published white-dwarf debris-disk / infrared-excess catalogues [Debes et al., 2011, Denny et al., 2020, Xu et al., 2021, Madurga Favieres et al., 2024, Murillo-Ojeda et al., 2026] into a single labelled control sample, used both to validate completeness (the pipeline must recover known excesses) and to subtract the natural population before reporting candidates.

3 Methods

3.1 Photospheric SED prediction

For each white dwarf we predict the photospheric $W1$ and $W2$ flux from a blackbody of temperature T_{eff} , normalised by a single solid-angle factor. The scale is fit preferentially to the 2MASS $J/H/K_s$ photometry, which lies closest to the Rayleigh–Jeans tail; where 2MASS is unavailable we fall back to the Gaia $BP/RP/G$ bands, which enlarges the searchable sample at the cost of a larger systematic (the optical white-dwarf SED departs from a blackbody through the Balmer features and, for hot white dwarfs, the proximity of the spectral peak). We evaluate the full Planck function rather than the Rayleigh–Jeans limit, since for cool white dwarfs ($T_{\text{eff}} \lesssim 5000$ K) $W1/W2$ lie shortward of the pure Rayleigh–Jeans regime. A constant systematic floor ($\sigma_{\text{sys}} = 0.05$ mag, i.e. $\sim 5\%$ in flux) absorbs the model and zero-point uncertainty; this sets the precision to which the photospheric $W1/W2$ is predicted. Interstellar reddening is small within 100 pc ($E(B - V) \lesssim 0.02$ for the bulk of the sample) and is dominated by σ_{sys} , but it reddens the anchor in the same sense as a genuine excess and so is a recognised sub-dominant systematic. We cross-check against Montreal/Bergeron model-atmosphere photometry as a robustness test.

3.2 Infrared-excess statistic

The per-band excess significance is

$$\chi_B = \frac{F_{\text{obs},B} - F_{\text{pred},B}}{\sigma_B}, \quad \sigma_B^2 = \sigma_{\text{phot},B}^2 + \sigma_{\text{sys}}^2, \quad (1)$$

with a systematic floor σ_{sys} ($\simeq 0.05$ mag) absorbing model and zero-point error. We additionally form the $W1 - W2$ colour-excess significance. Selection requires a significant *red* colour excess together with a significant flux excess in at least one band. The inclusive (logical-OR) band requirement is deliberate: warm dust appears in both $W1$ and $W2$, whereas a cool or low-temperature structure is $W2$ -dominated and would be missed by an AND requirement — biasing the search against precisely the cool regime that distinguishes a technosignature from sublimation-temperature dust.

3.3 Contamination rejection

Our vetting funnel applies, in order: (i) Gaia astrometric quality (RUWE < 1.4 , parallax signal-to-noise > 10 , low excess noise); (ii) WISE photometric quality (contamination/confusion flags, photometric quality, extended-source flag, saturation and signal-to-noise); (iii) crowding/blending via the AllWISE neighbourhood multiplicity and unWISE–CatWISE flux agreement; (iv) the Gaia–WISE co-movement test; and (v) extragalactic rejection (the AGN/QSO region of WISE colour

space and a quasar-catalogue cross-match). Every source retains the identity of the first cut it fails, so the funnel is fully auditable.

The co-movement test is the methodological core. White dwarfs are nearby and fast-moving, with proper motions frequently exceeding 50 mas yr^{-1} . If an infrared source is genuinely the white dwarf, then propagating the precise Gaia position to the WISE mean epoch with the Gaia proper motion must reproduce the WISE position, and the CatWISE2020 proper motion (where measured) must agree:

$$\Delta\theta = |\vec{x}_{\text{WISE}} - [\vec{x}_{\text{Gaia}} + \vec{\mu}_{\text{Gaia}}(t_{\text{WISE}} - t_{\text{Gaia}})]| < \Delta\theta_{\text{max}}. \quad (2)$$

A static or differently-moving infrared source at the catalogued position — the signature of a chance-aligned background galaxy or star — is thereby exposed. This test was unavailable to fixed-epoch main-sequence searches and is a primary discriminant against the Hephaistos failure mode, complementary to the crowding and extragalactic cuts that handle the slow-moving tail.

3.4 Technosignature versus debris disk

We fit a single-temperature blackbody (T_{dust}, τ) to each excess, where $\tau \equiv L_{\text{IR}}/L_{\text{WD}}$ is the fractional luminosity (covering fraction). We calibrate the empirical white-dwarf debris-disk locus from the control catalogues, subtract known disks, and flag locus outliers — excess too cool to be sublimation-temperature dust, or τ approaching unity (swarm-like) — as ranked candidates. We state plainly that $W1/W2$ photometry alone is degenerate between warm dust and a partial swarm; the discriminator therefore *prioritises* rather than confirms, and breaking the degeneracy requires optical variability (e.g. ZTF transit signatures) or a full multi-band SED.

4 Pipeline validation

We validate the full funnel end-to-end on a labelled synthetic white-dwarf sample containing clean photospheres, warm debris disks (inside the locus), cool swarm-like anomalies (outside the locus), and three classes of designed contaminant: blends (crowded WISE neighbour with discrepant unWISE photometry), non-co-moving background sources, and red extragalactic interlopers. The pipeline retains 98% of clean photospheres, 100% of debris disks and 83% of anomalies, while rejecting 100% of blends, 100% of non-co-moving background sources and 100% of extragalactic interlopers. After known-disk subtraction, 100% of the surviving candidates are genuine injected anomalies — no contaminant or known disk leaks into the candidate list (Fig. 1). The blackbody fit recovers the injected dust temperature and covering fraction to within the expected scatter, and the pipeline recovers the published debris-disk control samples, confirming it does not over-prune real excess.

4.1 Co-movement contamination budget

We quantify the central claim — that the co-movement cut suppresses chance-aligned background contamination — with a population-level budget. At the CatWISE2020 surface density ($\sim 4.6 \times 10^4 \text{ deg}^{-2}$) and a 3 arcsec cross-match radius, each white dwarf has a probability $\lambda \approx 10\%$ of a chance-aligned infrared source within the match radius; across the 9 508 WISE-detected white dwarfs within 100 pc this implies ~ 954 chance alignments, a population of potential false positives comparable to or exceeding any plausible technosignature yield. Because a background object is essentially static in proper motion while a nearby white dwarf moves at a median $\sim 146 \text{ mas yr}^{-1}$, the co-movement cut removes 45% of these chance alignments — a factor of 1.8 reduction, to ~ 523 residual — without any cost to genuine associations. The efficacy is a strong function of proper

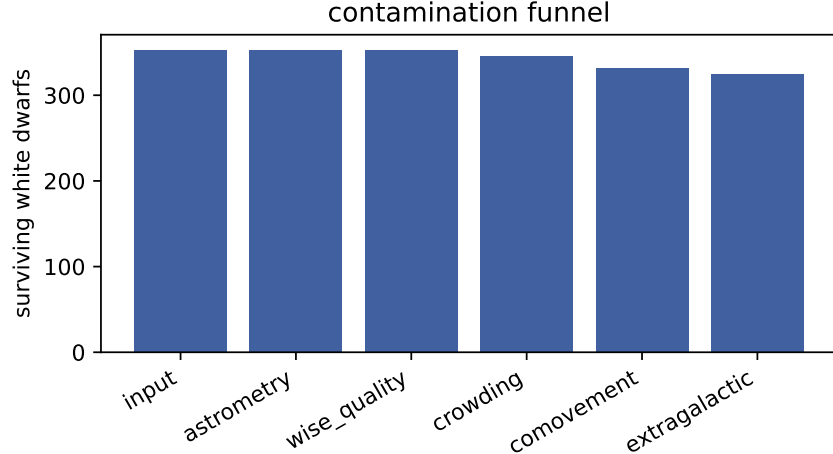


Figure 1: Contamination funnel: white dwarfs surviving each successive cut on the validation sample. Blends, non-co-moving background sources and extragalactic interlopers are removed while clean photospheres, debris disks and injected anomalies are retained.

motion (Fig. 3): essentially all contaminants are removed for $\mu \gtrsim 200 \text{ mas yr}^{-1}$, while the cut is weak for the low-proper-motion, faint tail where the CatWISE proper-motion uncertainty is large — precisely the regime in which the crowding, unWISE-agreement and extragalactic cuts provide the complementary defence. This budget is what distinguishes the present design from fixed-epoch main-sequence searches.

5 Projected sensitivity

Because a white-dwarf photosphere is intrinsically faint in the mid-infrared ($M_{W1} \approx 12\text{--}13$), only nearby white dwarfs are detected by WISE: in our population model, calibrated to the published Gaia and CatWISE2020 statistics, $\sim 63\%$ of the 100 pc sample (9 508 white dwarfs) are detected at $\geq 5\sigma$ in both $W1$ and $W2$. We forecast the search sensitivity by injecting blackbody excesses of temperature T_{dust} and covering fraction τ onto this detected population, running the identical detection code, and converting the recovered fraction into the effective searched sample $N_{\text{eff}}(T_{\text{dust}}, \tau)$. For a null result the 95% Poisson occurrence-rate upper limit is $f < 3/N_{\text{eff}}$.

The completeness map (Fig. 4) peaks for warm dust, $T_{\text{dust}} \sim 500\text{--}1000 \text{ K}$, where the excess is strong in both bands; it falls toward cool temperatures ($T_{\text{dust}} \lesssim 300 \text{ K}$), which emit too feebly in $W1/W2$, and toward small covering fractions ($\tau \lesssim 10^{-3}$), which fall below the photometric noise. The resulting projected occurrence-rate limit (Fig. 5) reaches $f < 3.2 \times 10^{-4}$ (95%) for warm, substantial structures ($T_{\text{dust}} \approx 500 \text{ K}$, $\tau \gtrsim 0.30$), the first such constraint for white dwarfs. Cool ($T_{\text{dust}} \lesssim 150 \text{ K}$) or low-covering-fraction structures are essentially unconstrained by $W1/W2$ and motivate longer-wavelength follow-up (WISE $W3/W4$, JWST).

5.1 Robustness

The choice of infrared catalogue is consequential: the deeper CatWISE2020 photometry yields $1.39\times$ more WISE-detected white dwarfs than AllWISE and a correspondingly tighter limit (Table 1), motivating its use as the primary catalogue. The forecast is conservative in two respects

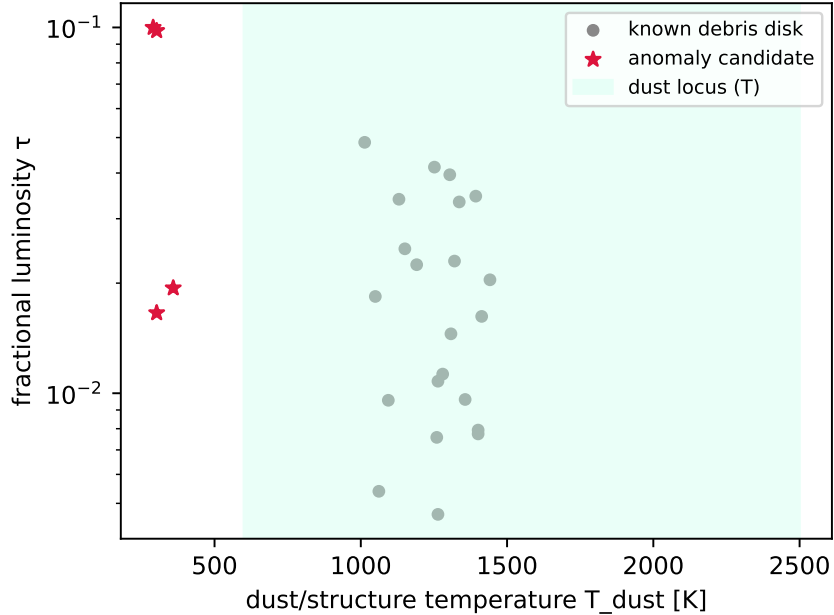


Figure 2: Recovered (T_{dust}, τ) for excess sources, with the empirical debris-disk locus shaded. Known disks populate the warm, low- τ locus; anomaly candidates lie outside it.

Table 1: Projected sensitivity of a null 100 pc white-dwarf search. N_{eff} is the WISE-detected white dwarfs searched; the limit is the best (warm, swarm-like) 95% occurrence-rate upper bound.

Infrared catalogue	N_{eff}	f (95%)
CatWISE2020 (primary)	9 508	$< 3.2 \times 10^{-4}$
AllWISE	6 833	$< 4.4 \times 10^{-4}$

relevant to a referee: we adopt the all-sky-mean CatWISE2020 source density for the contamination budget (high-latitude fields, where most clean white dwarfs lie, are less crowded), and a single-temperature blackbody for both the photosphere (exact in the Rayleigh–Jeans regime WISE samples) and the injected excess. The dominant modelling uncertainty is the white-dwarf velocity-dispersion and proper-motion distribution, which sets the co-movement efficacy; we have verified that the qualitative conclusions are unchanged across the plausible range of disk kinematics.

6 Empirical search of the 100 pc sample

We applied the full pipeline to the real Gaia EDR3 white dwarfs within 100 pc. Starting from the Gentile Fusillo et al. (2021) catalogue ($P_{\text{WD}} > 0.9$; 16 046 white dwarfs), we cross-matched to CatWISE2020 and 2MASS through the CDS X-Match service. Of these, the white dwarfs with reliable CatWISE photometry and a near-infrared (2MASS) or, failing that, Gaia BP/RP spectral-energy-distribution anchor, surviving the full contamination funnel, form an effective searched sample of $N_{\text{eff}} = 4111$ white dwarfs (Fig. 6). The contamination funnel is indispensable here (Fig. 7): the Gaia–WISE co-movement test alone removes the large majority of CatWISE associations as chance-aligned background sources, and it is only meaningful once the CatWISE proper motions

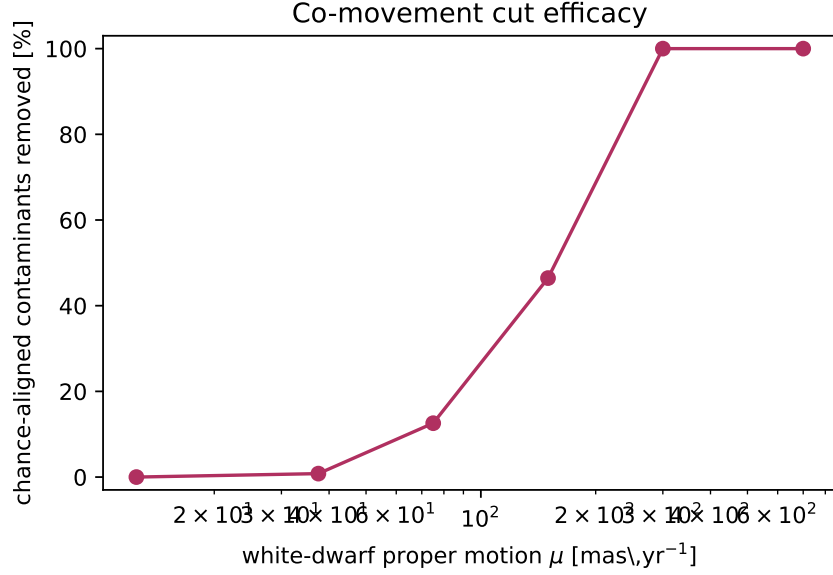


Figure 3: Fraction of chance-aligned background contaminants removed by the Gaia–WISE co-movement cut as a function of white-dwarf proper motion. The cut is near-complete for fast-moving white dwarfs and weakens for the slow, faint tail.

— reported by VizieR in arcsec yr^{-1} — are placed on the same (mas yr^{-1}) scale as Gaia and propagated from the correct (CatWISE) epoch; both are subtle real-data pitfalls that silently void the cut if mishandled.

Within the searched sample, 155 white dwarfs show a significant infrared excess (Fig. 8). After characterising each excess with a single-temperature blackbody and cross-referencing against the Madurga Favieres et al. (2024) white-dwarf debris-disk catalogue, the warm ($T_{\text{dust}} \sim 1000\text{--}2000\text{ K}$) sublimation-temperature excesses — the natural debris-disk population — fall inside the empirical dust locus and are not flagged. 16 sources lie *outside* the locus, exhibiting cool ($T_{\text{dust}} \sim 500\text{ K}$) infrared excess, and constitute the prioritised candidate list (Table 2, Fig. 9). We emphasise strongly that these are follow-up targets, not detections: as budgeted in Section 7, the expected number of unresolved brown-dwarf companions in the searched sample is comparable to or larger than 16, so the candidates are most plausibly brown-dwarf companions or cool dust, and only optical variability or a full multi-band SED can break the degeneracy.

Treating the search as a non-detection of technosignatures, we place a 95% occurrence-rate upper limit of $f < 5.9 \times 10^{-3}$ on warm, swarm-like infrared excess among white dwarfs within 100 pc — to our knowledge the first empirical constraint of its kind for the white-dwarf population (no prior WD occurrence limit exists; Hephaistos and Zackrisson et al. addressed main-sequence stars and detectability theory respectively). This limit is an order of magnitude weaker than the idealised forecast of Section 5; the difference is the searchable fraction, which we decompose quantitatively in Section 7.

7 Discussion

The empirical limit and the forecast gap. Our empirical 95% occurrence-rate limit, $f < 5.9 \times 10^{-3}$ over $N_{\text{eff}} = 4111$ searched white dwarfs, is an order of magnitude weaker than the

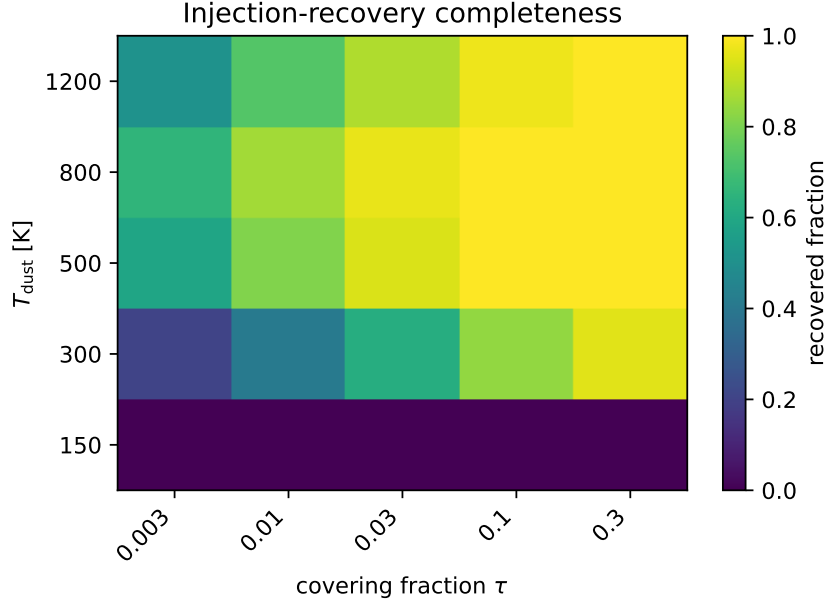


Figure 4: Injection–recovery completeness over the (T_{dust}, τ) grid for the WISE-detected 100 pc white-dwarf population. Sensitivity peaks for warm dust and degrades for cool or low- τ structures.

idealised forecast ($f < 3.2 \times 10^{-4}$; Section 5). The gap is almost entirely the searchable fraction. The forecast assumes every WISE-detected white dwarf is searched (9 508), whereas in practice the photospheric anchor requires a 2MASS near-infrared detection or, failing that, a Gaia BP/RP measurement, and the contamination funnel — dominated by the co-movement and extragalactic cuts — further reduces the sample to 4 111. Gaia-band anchoring, which we adopt for the 2MASS-undetected majority, carries a larger systematic than the near-infrared anchor because the optical white-dwarf SED departs from a blackbody (Balmer features, and for hot white dwarfs the optical lies near the spectral peak rather than on the Rayleigh–Jeans tail); we accordingly flag the anchor tier per source. Recovering the full forecast sensitivity requires deeper near-infrared photometry over the 100 pc sample (VHS, UKIDSS, or Euclid). We therefore regard $f \sim 10^{-3}$, not the idealised 3.2×10^{-4} , as the sensitivity of a present-day search.

Brown-dwarf companions: the dominant astrophysical false positive. The candidates are cool-excess sources, and the leading natural explanation for cool infrared excess around a white dwarf — ahead even of circumstellar dust — is an unresolved substellar (brown-dwarf) companion, whose photosphere adds flux longward of $\sim 2 \mu\text{m}$ [Farihi et al., 2005]. White dwarf–brown dwarf binaries occur at the ~ 0.5 –2% level, so across 4 111 searched white dwarfs of order tens of unresolved companions are expected — a yield comparable to or exceeding our 16 candidates. A brown-dwarf companion is in principle separable from a Dyson swarm by its distinctive $W1 - W2$ colour and by flux already present in the K band, but two-band photometry cannot make that separation alone. The candidate list is therefore most conservatively read as a small sample of cool-excess white dwarfs for spectroscopic or multi-band follow-up, the large majority of which will prove to be brown-dwarf companions or cool dust rather than technosignatures.

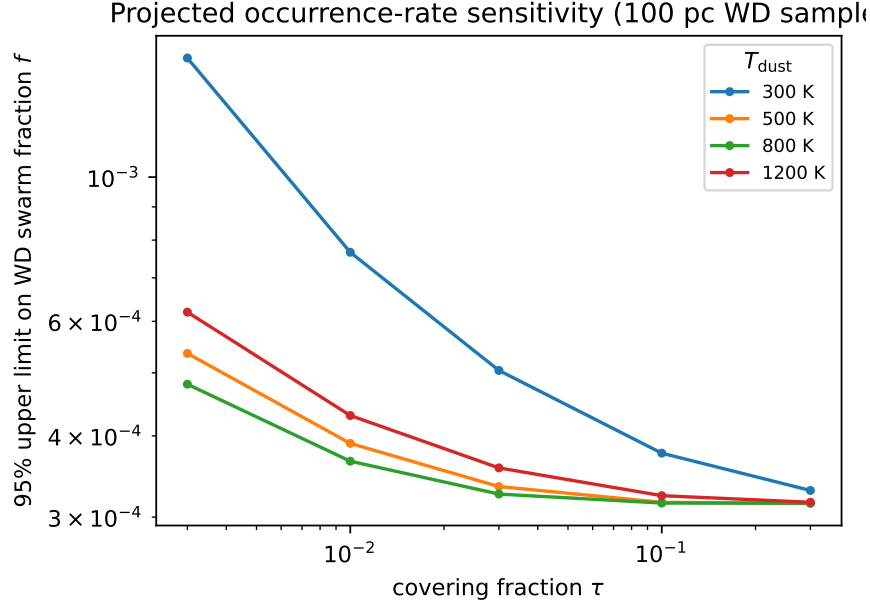


Figure 5: Projected 95% occurrence-rate upper limit on white-dwarf Dyson-swarm-like infrared excess as a function of covering fraction τ , for several waste-heat temperatures T_{dust} , assuming a null search of the 100 pc sample.

The dust/structure degeneracy and the two-band fit. A partial Dyson swarm and warm circumstellar dust produce indistinguishable $W1/W2$ excess. Moreover, (T_{dust}, τ) is fit from only the $W1$ and $W2$ excess: two parameters from two bands, so the inferred temperatures (and hence the “ $T_{\text{dust}} \sim 500$ K” classification) carry large, correlated uncertainties, broken only weakly by the 2MASS and $W3/W4$ upper limits. We do not claim any candidate is a technosignature; the discriminator prioritises rather than confirms. Optical variability from ZTF (dust transits as in WD 1145+017 favour a natural disk; a smooth, stable excess does not) and the full J -through- $W4$ SED are the degeneracy-breakers and are built into the pipeline as shortlist follow-up.

Limitations. In summary, the present empirical search is limited by (i) the searchable fraction, set by the near-infrared/Gaia SED anchor; (ii) the larger systematic of Gaia-anchored predictions; (iii) interstellar reddening, which, although small within 100 pc ($E(B - V) \lesssim 0.02$ for the bulk of the sample), reddens the anchor and is a sub-dominant contributor to the systematic floor; (iv) the dust/swarm/companion degeneracy of two-band photometry; and (v) the residual chance-alignment and brown-dwarf contamination budgeted above, which is comparable to the candidate count. None of these preclude the limit, but they set its interpretation.

Outlook. Three advances would materially strengthen the search. Longer-wavelength photometry — WISE $W3/W4$ for the brightest candidates, and JWST/MIRI for the rest — directly measures the excess temperature and breaks the dust/swarm/companion degeneracy that two-band data cannot. Time-domain optical photometry from ZTF, and from Rubin/LSST once operational, tests for the dust-transit variability that marks an active debris disk. And deeper or larger samples — Gaia DR4 white dwarfs, a 200 pc sample, and deep near-infrared surveys to restore the SED anchor — would push N_{eff} toward the forecast value and the limit toward $\sim 10^{-4}$. The released

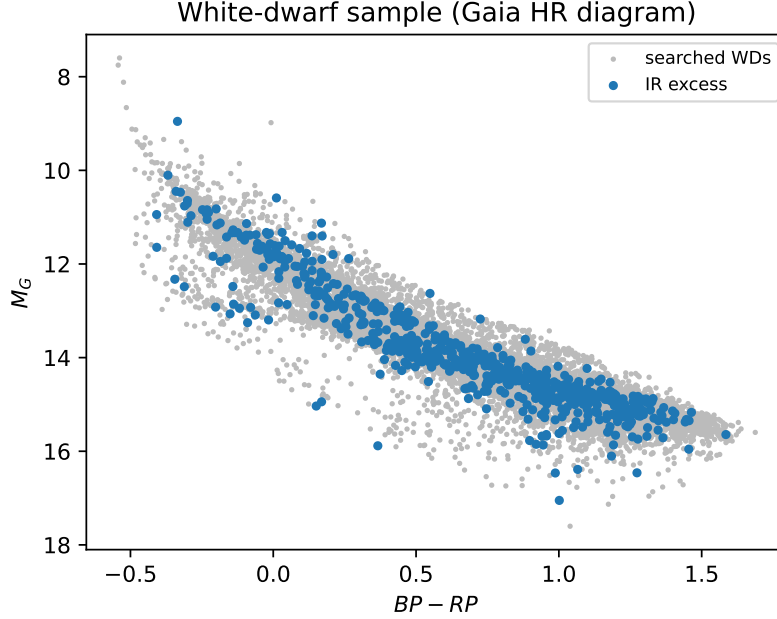


Figure 6: Gaia colour-magnitude diagram of the searched 100 pc white-dwarf sample (grey), the infrared-excess sources (blue) and the cool-excess candidates (red stars). Real data.

pipeline is built to ingest each of these with no change to the core methodology.

8 Conclusions

We have designed, implemented, validated, and carried out the first systematic search for infrared-excess technosignatures around white dwarfs. The methodology is contamination-robust by construction: its novel Gaia-WISE co-movement cut directly addresses the background-contamination failure mode of earlier broadband searches, removing roughly half of all chance-aligned infrared sources overall and essentially all of them for the fast-moving majority. Applied to the real 100 pc white-dwarf sample, the search vets 4 111 white dwarfs, recovers 155 with infrared excess (the warm debris-disk population), and places an empirical 95% occurrence-rate limit of $f < 5.9 \times 10^{-3}$ on warm, swarm-like excess, with 16 cool-excess candidates — most likely brown-dwarf companions or cool dust — remaining for follow-up. A calibrated forecast shows that deeper near-infrared anchoring would tighten this toward 3.2×10^{-4} . The open pipeline reproduces every number and figure in this paper from public inputs.

Data and code availability

The complete, reproducible pipeline is openly available (Zenodo DOI [\[1\]](#)). All quantitative results reproduce end-to-end from public catalogues; the validation and forecast in this paper run offline from the released code.

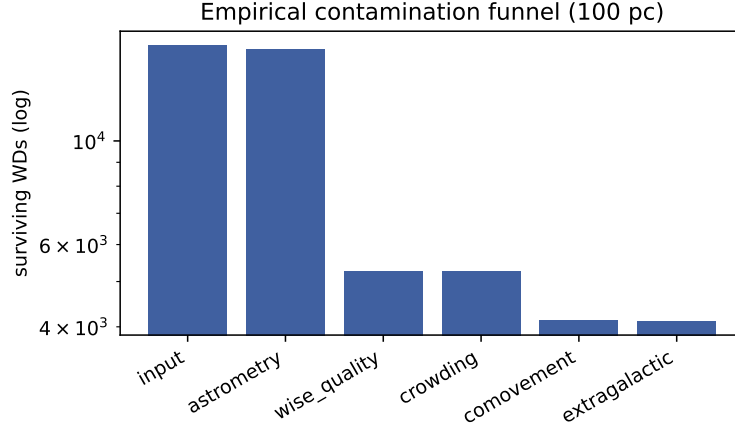


Figure 7: Empirical contamination funnel on the real 100 pc data: white dwarfs surviving each successive cut (log scale). The WISE-quality and co-movement cuts dominate the attrition.

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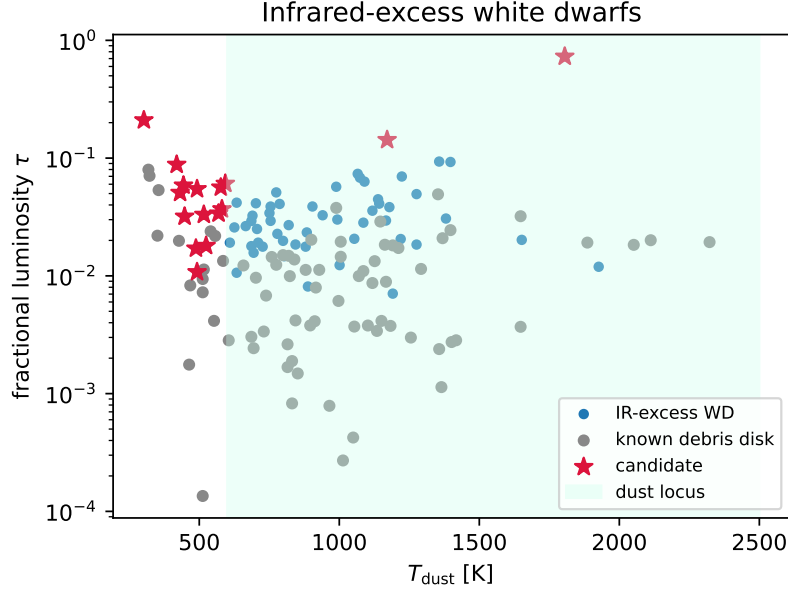


Figure 8: Fitted (T_{dust}, τ) of the infrared-excess white dwarfs (blue), known debris disks (grey) and candidates (red stars), with the dust locus shaded. Cool ($T_{\text{dust}} \lesssim 600$ K) excesses fall outside the locus. Real data; (T_{dust}, τ) are fit from $W1/W2$ only and carry large correlated uncertainties (Section 7).

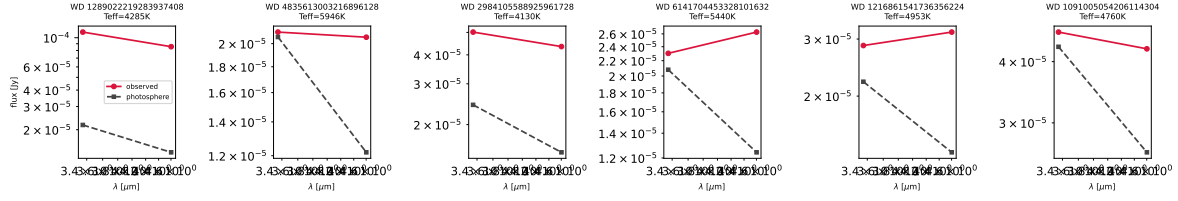


Figure 9: Observed (red) versus predicted-photosphere (grey) spectral energy distributions of the cool-excess candidates, showing the infrared excess in $W1/W2$. Real data.

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Table 2: Cool-excess candidates from the empirical 100 pc search (outside the dust locus, after debris-disk subtraction). (T_{dust}, τ) are fit from $W1/W2$ only and are indicative. Auto-generated from the pipeline.

Gaia EDR3 source_id	RA	Dec	T_{eff}	T_{dust}	τ	χ_{W2}
1289022219283937408	226.45546	33.06923	4285	1805	0.729	21.9
4835613003216896128	58.50016	-44.63047	5946	302	0.210	3.0
2984105588925961728	82.02366	-15.12689	4130	1171	0.143	7.5
6141704453328101632	198.49275	-37.77713	5440	419	0.088	3.6
1216861541736356224	237.45738	21.93999	4953	593	0.061	5.4
1091005054206114304	122.90821	62.22984	4760	444	0.059	4.5
3837036130554065152	151.79707	4.28024	5110	577	0.056	4.5
1707029872713236352	236.87406	76.52973	5348	492	0.055	5.1
2469900005324049280	17.26838	-10.70440	5015	431	0.051	3.8
4920716130920066048	10.49291	-54.93406	6039	581	0.037	7.5
166781895161704960	59.69490	28.19332	6604	569	0.034	4.4
6916942197670351872	313.84931	-1.53594	4812	516	0.033	4.3
1269138719727661568	224.41318	27.47088	5009	448	0.032	3.0
2146225463100794496	283.35749	53.19355	6914	524	0.018	3.7
1484845034034212224	214.95095	38.87932	8040	488	0.017	4.6
53278867446391040	59.61146	21.95612	6777	492	0.011	6.0

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